

11. (2 points)

Given the following information as of December 31, 2018:

Accident Year	Incurred Losses (\$000)	Chain Ladder Ultimate (\$000)	Bornhuetter-Ferguson Ultimate (\$000)	Bayesian Ultimate (\$000)
2013	4,400	4,400	4,400	4,400
2014	5,000	6,000	5,833	5,840
2015	2,800	4,000	4,300	4,278
2016	3,000	5,400	5,222	5,301
2017	4,000	10,000	7,000	7,750
2018	750	3,000	4,500	

- $\alpha_{2018} = 12,500$
- $\beta_{2018} = 2.5$
- $\phi_{2018} = 1.1$

a. (0.5 point)

Compare the variance of the prior distributions for accident years 2016 and 2017.

b. (0.5 point)

Calculate the Bayesian ultimate loss estimate for accident year 2018.

c. (0.5 point)

Calculate the accident year 2014 correlated chain ladder log mean ultimate loss given the following parameters:

- $\alpha_{2013} = 8.5172$
- $\alpha_{2014} = 8.5172$
- $\beta_6 = 0$
- $\rho = 0.2$

d. (0.5 point)

Briefly describe one advantage of each of the following methods:

- Correlated chain ladder method
- Bayesian method

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QUESTION 11	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE(S): A5, A6, A8, A10
SAMPLE ANSWERS	
Part a: 0.5 point	
<u>Sample 1</u> AY 2016: Bayesian estimate is slightly closer to BF method than C-L. Variance is moderately small, but not too small AY 2017: Bayesian estimate is very close to BF method. Thus, strong opinion, small variance The variance for AY 2016 is larger than AY 2017 <u>Sample 2</u> 2016: $5301 = 5400 * Z + 5222 * (1 - Z) \Rightarrow Z = 44.4\%$ 2017: $7750 = 10,000 * Z + 7000 * (1 - Z) \Rightarrow Z = 25\%$ Since Z is smaller for 2017, β is bigger, thus the variance (α/β^2) is smaller in 2017 <u>Sample 3</u> AY 2016: $Z*(5400) = (1-Z)*5222 = 5301$ $Z = .4438$ $p_{2016} = 3000/5400 = .5556$ $.5556/[\beta\phi + .5556] = .4438$ $\beta\phi = .696$ AY 2017: $Z*(10,000) + (1-Z)*(7000) = 7,750$ $Z = .25$ $p_{2017} = 4000/10000 = .4$ $.4/[\beta\phi + .4] = .25$ $\beta\phi = 1.2$ Assuming the dispersion parameter used in 2016 is not significantly smaller than that used in AY 2017, we can assume a higher β value for AY 17 & thus a lower variance of the prior distribution in AY 17	
Part b: 0.5 point	
$p_k = 750/3000 = 0.25$ $Z = .25/ [.25 + 1.1*2.5] = 0.083$ $0.083*3000 + (1 - 0.083)*4500 = 4376$	
Part c: 0.5 point	
<u>Sample 1</u> $\mu_{2013} = 8.5172 + 0 = 8.5172$ $\mu_{2014} = 8.5172 + 0 + 0.2 (\ln(4400) - 8.5172)$ $= 8.4916 \quad (ult = e^{8.4916} = 4874)$	

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<u>Sample 2 [assumed different scale was used for α and θ]</u> $\mu_{2013} = 8.5172 + 0 = 8.5172$ $\mu_{2014} = 8.5172 + 0 + 0.2 (\ln(4,400,000) - 8.5172)$ $= 9.873$
Part d: 0.5 point
<u>Sample Responses for Correlated Chain Ladder advantages</u> • Incorporates correlation between AYs • Shown by Meyers to accurately describe the distribution of reserves <u>Sample Responses for Bayesian advantages</u> • Can incorporate expert opinion to the simulation • Can produce full predictive distribution in addition to point estimate
EXAMINER'S REPORT
Candidates were expected to know the definition, assumptions, calculation procedures and advantages for the Bayesian and Correlated Chain Ladder reserving methods.
Part a Candidates were expected to know that when the distribution of initial expected ultimate has a high variance, the Bayesian ultimate will be closer to the chain ladder ultimate and then apply that to compare the variances of the distributions of initial expected ultimate losses between accident years. Alternatively, candidates could calculate the variances of the prior distributions and compare. Common mistakes included: • Stating that when the distribution of initial expected ultimate losses has a high variance, then the Bayesian ultimate will be closer to the Bornhuetter-Ferguson (B-F) ultimate. • Defining closeness based on subtracting Bayesian from B-F and not as a ratio to the range of Chain ladder to B-F
Part b Candidates were expected to be able to calculate the Bayesian ultimate loss estimate. Common mistakes included: • Using an incorrect formula for calculating credibility • Using expected ultimate claims as the complement of credibility instead of the B-F estimate
Part c Candidates were expected to be able to calculate the logged mean ultimate of the Correlated Chain Ladder model using given parameters. Common mistakes included: • Using an incorrect equation

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Part d

Candidates were expected to know advantages of the Correlated Chain Ladder and Bayesian methods.

Common mistakes for CCL included:

- Correlated Calendar years
- Incorporate calendar year effects (trend, claim settlement rate – see CIT and CSR models)
- Passes the K-S test

Common mistakes for Bayesian included:

- It credibility weights chain ladder and Bornhuetter-Ferguson
- Minimizes MSE
- You can assume an a priori distribution
- Adds stability